
Claim-status ledger - Paper X

Every load-bearing claim, the exact rung it sits on, and where the evidence is. Nothing here may be read one rung higher than it is placed. Rungs: **P** proved-within-model · **V** internally-verified (code matches maths) · **S** synthetically-validated · **Pilot** real-model pilot (limited, provisional) · **Open** open empirical problem · **Hyp** falsifiable hypothesis · **Cond** conditional implication.

#	Claim	Rung	Evidence
1	Stability of recursive self-improvement is governed by the exponent inequality $\beta > k$ (correction must out-scale drift-acceleration).	P	Theorems 1, 3, 4; §3.4-3.8
2	Exact steady-state misalignment fraction $d^* = \gamma_1 r / (A+r)$, reducing to $\rho = \gamma_1 r / A$ when $A \gg r$.	P	Theorem 1; §3.5
3	In the gain-only model the fraction saturates at γ_1, it does not diverge to infinity (corrects the prior draft).	P	Theorem 2; §3.4
4	A finite-time hard takeoff is alignment-stable iff $\beta > k$; the verdict $= \mathrm{sign}(\beta - k)$, independent of speed and of the singularity time.	P	Theorem 3 (Hard-Takeoff Depth-Regularity); §3.7
5	Genuine divergence lives in a distinct compounding channel with threshold $\rho_{\mathrm{prop}} = (\gamma_3 - 1)r/A < 1$.	P	Theorem 4; §3.8
6	Vector misalignment: an exact spectral-abscissa threshold; misalignment persists on the corrector's null subspace ("you cannot correct what you do not measure").	P	Theorem 5; §3.9
7	Stochastic drift: an Ornstein-Uhlenbeck tail bound for governance.	P	Theorem 6; §3.10
8	The threshold shares the form of the quantum error-correction sub-threshold condition - not its mechanism (QEC suppression is exponential; this is power-law).	Hyp (falsifiable, F4)	§3.12; F4; abstract disanalogy
9	The numerical implementation matches the closed-form derivation, and the corrected theorems are independently re-verified from scratch (Thm 2 time-varying limits; Thm 4 equality = linear divergence; Thm 5 non-normal transient + null-axis floor; Thm 6 OU; $\gamma_3=1$ boundary).	V	code/experiment_coscaling.py; results/verdicts.json; test_coscaling.py (12); test_theorems_independent.py (14, no harness import)
10	The β/k estimator recovers known exponents from model-generated data (to ≈ 0.1) and fails honestly on saturated data.	S	experiments/scripts/estimate_exponents.py; results/realmodel/exponent_estimates.json
11	On a real model (Claude), the external corrector removes a seeded reward-hack: blind D : 10 to 0\$, capability C : 0 to 1.0\$.	Pilot (mechanism; IVd non-compliant - same-family scoring)	results/realmodel/claude-opus_corrector_probe.json; REAL_MODEL_CLAUDE_RESULTS.md
12	The co-scaling dynamic (H_1 decoupled drifts up; H_2 coupling bounds d) on a real model.	NOT shown (null contrast; the model did not drift)	results/realmodel/claude-opus_2026*.json (H1=false, H2=false)
13	β and k measured on a real drifting self-improving system across capability levels.	Open	experiments/PROTOCOL.md (the named next experiment)

#	Claim	Rung	Evidence
14	If 13 succeeds, $\beta - k$ becomes a measurable corrigibility audit margin for governance.	Cond	§9 (stated conditionally)
15	$\beta > k$ holds only in the unbounded power-law corrector regime; a finite-capacity corrector ($A \rightarrow A_{\max}$) fails it asymptotically for $k > 0$ (since $r \rightarrow \infty$ outpaces a fixed $A_{\max}$).	P (scope limit)	§8 "Unbounded vs finite-capacity corrector"; Theorem 4
16	$d = D/C$ is an operational, scoring-convention-dependent risk index, not a natural constant; the criterion is in scale-free exponents.	P (scope)	§8 "Metric normalisation"

Priority vs novelty (separate axes)

- **Priority** (when the author articulated it): the thesis was set out in *Infinite Architects* (copyright deposited 8 Dec 2024; published 2 Jan 2026; ISBN 978-1806056200), with each paper time-stamped on OSF (10.17605/OSF.IO/6C5XB). This is real and is the author's.
- **Novelty** (new to the literature): the co-scaling *intuition* has ancestors (Ashby 1956; Conant-Ashby 1970; scalable oversight; Lyapunov drift), credited in §2. The defensible *novel* residue is: the closed-form $\beta > k$ criterion + the Hard-Takeoff Depth-Regularity theorem (this paper); the IV.d blinding-reversal measurement law (companion); and the runnable β/k estimator + protocol. Programme-wide prior-art audits are committed under each paper's results/.

What would falsify or materially weaken the central claim

- A stable system with $\beta < k$ under genuine acceleration, or an unstable one with $\beta > k$ (kill conditions F1-F3', F6 in the stated model).
- A real drifting system whose measured (β, k) does **not** predict whether coupled correction succeeds or fails.
- The QEC correspondence (claim 8) is downgraded - not the threshold result - if the suppression is shown to be exponential rather than power-law (F4).